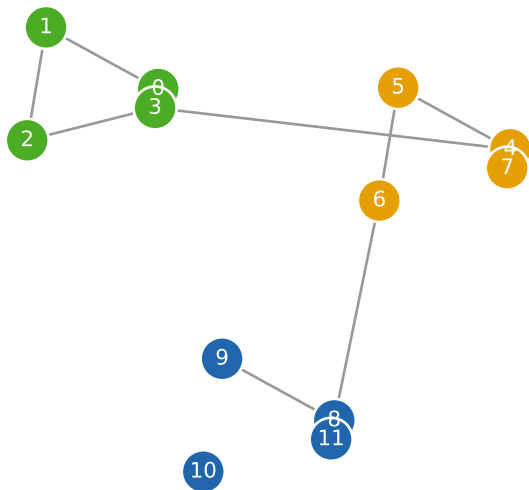
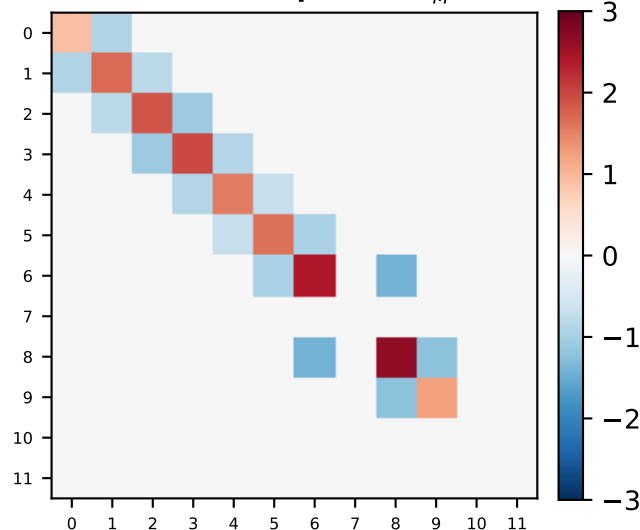


Figure S1: Worked toy example — 12 metabolites, 8 reactions, full pipeline with real computed values (CH-PSI)

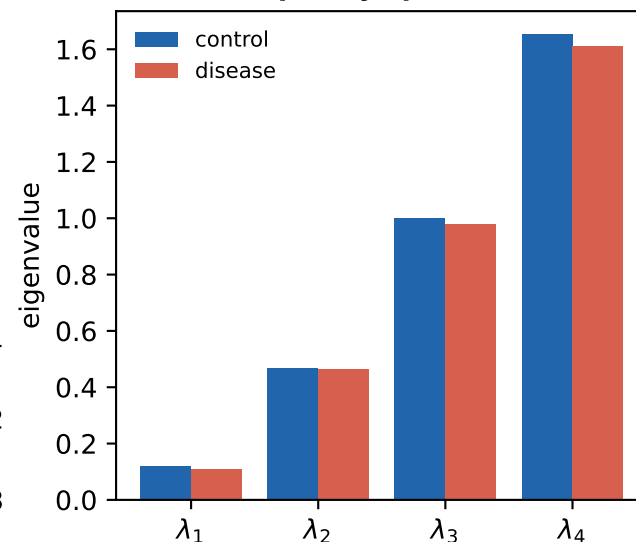
A Toy network (12 metabolites, 8 reactions)



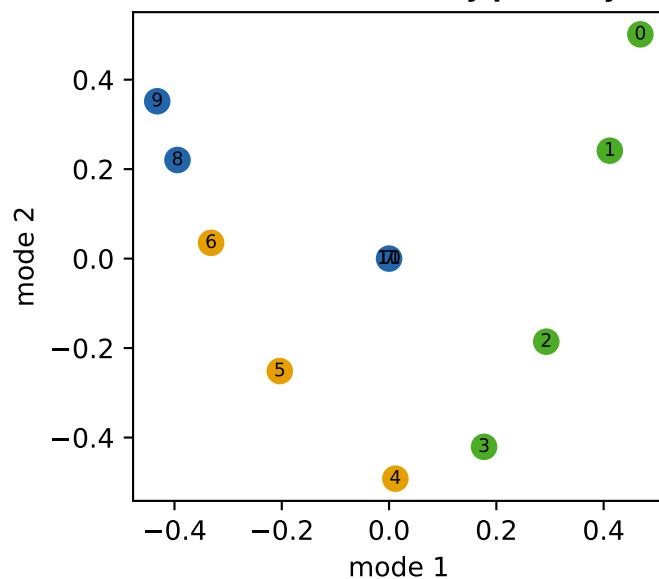
B Metabolite Laplacian $\Delta_M^{(c)}$ (disease)



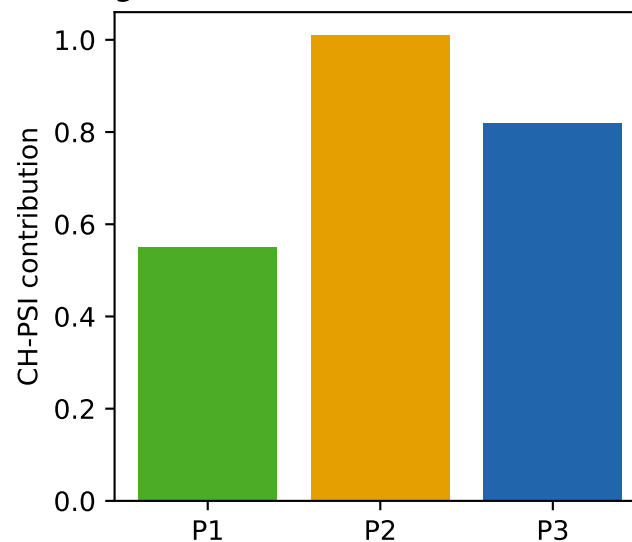
C Low-frequency spectrum



D Spectral embedding (metabolites cluster by pathway)



E Pathway separability (CH-PSI) global = 3.26



F Panel selection (size 6) geometry-aware beats 98% of random

